

## EXPERIMENT NO. 15:

### Implement Iris Flower Classification using Naive Bayes Classifier

#### Aim:

To implement Iris Flower Classification using the Naive Bayes Classifier in Python and classify a new Iris flower sample.

#### Procedure:

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Separate the input features and target classes.
4. Split the dataset into training and testing sets.
5. Create a Gaussian Naive Bayes classifier.
6. Train the model using the training dataset.
7. Predict the class labels for the test data.
8. Calculate and display the classification accuracy.
9. Predict the species of a new Iris flower sample.

#### Program:

```
# Iris Flower Classification using Naive Bayes

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score

# Load Iris Dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

# Create Naive Bayes Classifier
model = GaussianNB()
```

```
# Train the model

model.fit(X_train, y_train)

# Predict test data

y_pred = model.predict(X_test)

# Calculate accuracy

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:")

print(round(accuracy * 100, 2), "%")

# Predict a new Iris flower

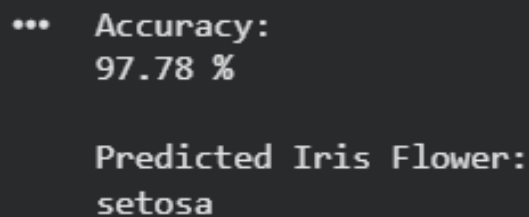
new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_flower)

print("\nPredicted Iris Flower:")

print(iris.target_names[prediction][0])
```

Output:



```
*** Accuracy:
    97.78 %

    Predicted Iris Flower:
    setosa
```

Result:

The Iris Flower Classification program was successfully implemented using the Naive Bayes Classifier. The model classified the new Iris flower sample as Setosa and achieved an accuracy of approximately 97.78%.